

Advanced (Habanero)

- Q1.** What is the graph of the inequality $x + 2y > 4$?
(A) A solid line with a shaded region above it
(B) A dashed line with a shaded region below it
(C) A solid line with a shaded region below it
(D) A dashed line with a shaded region above it
(E) A vertical line with a shaded region to the left
- Q2.** Which of the following inequalities has a solid boundary line?
(A) $x - 2y \geq 3$
(B) $x + 3y < 2$
(C) $2x - 4y > 1$
(D) $3x + 2y \leq 5$
(E) $x - y = 2$
- Q3.** What is the solution region for the inequality $y < -2x + 1$?
(A) The region below the line $y = -2x + 1$
(B) The region above the line $y = -2x + 1$
(C) The region to the left of the line $y = -2x + 1$
(D) The region to the right of the line $y = -2x + 1$
(E) The region on the line $y = -2x + 1$
- Q4.** Which inequality represents the shaded region?
(A) $x + y > 2$
(B) $x - y < 1$
(C) $2x + 3y \geq 5$
(D) $y - 2x < 3$
(E) $x + 2y \leq 4$
- Q5.** What type of line is used to graph the inequality $x \geq 2$?
(A) Solid vertical line
(B) Dashed vertical line
(C) Solid horizontal line
(D) Dashed horizontal line
(E) Slanted line
- Q6.** The inequality $y > x$ represents which type of region?
(A) The region above the line $y = x$
(B) The region below the line $y = x$
(C) The region to the left of the line $y = x$
(D) The region to the right of the line $y = x$
(E) The region on the line $y = x$
- Q7.** The inequality $x < 3$ represents which type of region?
(A) The region to the left of the line $x = 3$
(B) The region to the right of the line $x = 3$
(C) The region above the line $x = 3$
(D) The region below the line $x = 3$
(E) The region on the line $x = 3$
- Q8.** The inequality $y \leq -x + 2$ has which type of boundary line?
(A) Solid line
(B) Dashed line
(C) Vertical line
(D) Horizontal line
(E) Slanted line

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $x + 2y > 3$ and identify the solution region.

- Q10.** Write an inequality that represents the shaded region and graph it.

Chef's Tips

- Tip 1: Read the questions carefully and choose the correct answer.
- Tip 2: Use a graphing tool to help visualize the inequalities.
- Tip 3: Check your work to ensure accuracy.